CUSTOMER INTELLIGENCE SYSTEM

Complete Architecture, File Breakdown & Execution Guide

CLIENT REFERENCE V2.0

1. PROJECT KA MAQSAD (WHAT IS THIS PROJECT?)

Customer Behavior & Feedback Intelligence System aik advance full-stack web application hai jo e-commerce store ke customer ke **har action** ko track karti hai, unka feedback leti hai, aur **Google Gemini AI** ki madad se deep business insights aur customer behavior patterns generate karti hai.

🛒 Customer Facing Store

Customers products browse kar saktay hain, live search kar saktay hain, cart main add kar saktay hain, feedback de saktay hain, aur complete **Checkout demo** kar saktay hain.

📊 Admin & Intelligence Hub

Admin tamam feedback, live customer activity, rating charts, complete customer sessions (Customer 360), aur **Gemini AI automated summaries** dekh sakta hai.

2. KONSI TECHNOLOGY USE HUI AUR KYUN? (TECH STACK & JUSTIFICATION)

TECHNOLOGY	ROLE IN PROJECT	KYUN USE KI? (WHY THIS TECH?)
`React 19 + Vite`	Frontend UI Engine	Fastest build speeds, zero lag, reactive component updates, single-page seamless experience.
`Tailwind CSS v4`	Design & Styling	Modern monochrome luxury editorial theme (#fcfcfc), responsive layouts, clean spacing without bulky CSS files.
`Zustand`	Client State Management	Redux se bohat halka (lightweight), zero boilerplate, cart aur authentication persist storage ke sath instantly handle karta hai.
`Recharts`	Interactive Visualizations	Rating distributions, sentiment breakdown, page views, aur Engagement vs Sentiment scatter plots render karne ke liye.
`Node.js + Express 5`	Backend API Server	High throughput asynchronous event handling, structured REST API routing, secure request authentication.
`MongoDB + Mongoose`	Database Engine	Flexible schema design — user behavior events, metadata maps, feedbacks aur order receipts ko seamlessly store karne ke liye.
`Google Gemini AI`	AI Insights & Summaries	Raw customer feedback aur behavior data ko analyze karke automated executive summaries aur business recommendations deta hai.
`JWT + Bcrypt`	Security & Auth	Industry-standard password encryption aur stateless Bearer Token authorization for Customer & Admin roles.

3. PROJECT RUN KARNE KA ASAAN TAREEQA (STEP-BY-STEP SETUP)

1. **Prerequisites Check**

   Ensure karein ke aapke computer main
   Node.js (v18+)
   aur
   MongoDB
   (locally ya MongoDB Atlas) install aur running hain.

2. **Backend Server Setup & Seed Data**

   Pehle server folder main ja kar dependencies install karein aur initial data seed karein:

   ```
   cd server
   npm install
   node seed/seedData.js # (Creates initial realistic products, demo customers & feedback)
   ```

3. **Start Backend Server**

   Terminal main server start karein (Port 5000 par chalega):

   ```
   npm run dev
   ```

4. **Frontend Setup & Run**

   Naye terminal tab main client folder open karein aur run karein (Port 5173 par chalega):

   ```
   cd client
   npm install
   npm run dev
   ```

5. **Access Application**

   Browser main
   http://localhost:5173
   open karein.

FOLDER & FILE STRUCTURE GUIDE

Har folder aur file ka asaan explanation aur purpose

CODEBASE ANATOMY

4. SERVER DIRECTORY BREAKDOWN (BACKEND)

FOLDER / FILE	ISKA KAAM KYA HAI? (PURPOSE & RESPONSIBILITY)
`server.js`	Backend ka main entry point. Express app create karta hai, MongoDB connect karta hai, aur saray API routes register karta hai.
`models/User.js`	User database schema (Name, Email, Bcrypt Hashed Password, Role: 'customer' ya 'admin').
`models/Product.js`	Product schema (Name, Description, Category, Price, Image URL, Rating).
`models/Feedback.js`	Customer feedback schema (Rating 1–5, Comments, Category, Sentiment: Positive/Neutral/Negative).
`models/BehaviorEvent.js`	Live behavior tracking schema (page_view, product_view, product_click, add_to_cart, checkout_completed, duration).
`models/Order.js`	Checkout database schema (Order Number, Customer info, Shipping address, Items array, Total, Payment status).
`controllers/authController.js`	User Registration aur Login logic, credentials verify karke JWT authentication token generate karta hai.
`controllers/productController.js`	Database se products fetch karna, filter karna, aur specific product details laane ka logic.
`controllers/behaviorController.js`	Frontend se aane walay live tracking events aur user time duration ko database main store karta hai.
`controllers/analyticsController.js`	Admin dashboard stats aggregate karta hai, Customer 360 journey merge karta hai, aur scatter plot data banata hai.
`controllers/orderController.js`	Orders place karta hai, database main save karta hai, aur automatic checkout behavior event trigger karta hai.
`controllers/aiController.js`	Gemini AI API ko call karke feedback summary aur business recommendations mangwata hai.
`routes/`	API URL endpoints define karta hai (`/api/auth` , `/api/products` , `/api/orders` , etc.).
`middleware/auth.js`	JWT Token verify karta hai taake protected routes sirf authorized users ya admins hi open kar saken.
`services/aiService.js`	Google Gemini API SDK integration. Fallback rule-based sentiment engine bhi include hai agar internet/quota na ho.
`seed/seedData.js`	Database ko real products, customers, aur feedbacks se fill karne ke liye automated script.

5. CLIENT DIRECTORY BREAKDOWN (FRONTEND)

FOLDER / FILE	ISKA KAAM KYA HAI? (PURPOSE & RESPONSIBILITY)
`src/App.jsx`	Frontend ka central routing hub. Saray pages (Store, Product, Checkout, Admin, Customer 360) manage karta hai.
`src/pages/Products.jsx`	Main shopping catalog page jahan saray products grid main show hotay hain with search bar aur cart trigger.
`src/pages/ProductDetails.jsx`	Single product ka full detail page with add-to-bag, quantity selector, accordion details, aur feedback button.
`src/pages/Checkout.jsx`	Dedicated checkout screen. Address form, delivery method, demo card details, promo code, aur place order button.
`src/pages/OrderConfirmation.jsx`	Order place hone ke baad ka receipt page with unique Order Number, items list, financial breakdown, aur print receipt option.
`src/pages/AdminDashboard.jsx`	Complete Intelligence Dashboard. Stats cards, Recharts graphs, Feedback database, Orders table, aur Gemini AI cards.
`src/pages/CustomerTimeline.jsx`	Customer 360 Journey. Specific user ka session-by-session timeline (Page views, clicks, cart adds, feedback, aur orders).
`src/components/CartSidebar.jsx`	Slide-over shopping bag drawer jo subtotal dikhata hai aur Proceed to Checkout navigate karta hai.
`src/components/FeedbackModal.jsx`	Popup modal jahan customer star rating (1–5) aur feedback comment submit karta hai.
`src/components/ProtectedRoute.jsx`	Security wrapper jo non-admin users ko Admin Dashboard access karne se rokta hai.
`src/utils/behaviorTracker.js`	Automated engine jo customer ke page views, product clicks, aur time duration ko background main capture karta hai.
`src/store/`	Zustand stores (`authStore.js` for user/token, `cartStore.js` for bag items).
`src/services/api.js`	Axios HTTP client with automatic Authorization token interceptor.

DEMO WORKFLOW & EXAMINER QUICK REFERENCE

Client ya Examiner ko project dikhane ka step-by-step tareeqa

PRESENTATION GUIDE

6. LOGIN CREDENTIALS (DEMO ACCOUNTS)

🔑 Admin Account (Full Dashboard Access)

Email: `admin@example.com`
Password: `password123`
URL: `http://localhost:5173/login`
Note: Login karne ke baad top navbar main "Admin" ka button aa jayega.

👤 Customer Account (Shopping & Feedback)

Naya customer register kar saktay hain on `/register` ya guest checkout use kar saktay hain. Har logged-in customer ka journey alag se Customer 360 timeline main track hota hai.

7. STEP-BY-STEP PRESENTATION FLOW (CLIENT/EXAMINER DEMO)

Step 1 — Customer Shopping & Behavior Tracking

1. `/products` open karein aur product browse karein.
2. Product par click karein (view event record ho jayega).
3. "Add to Bag" click karein aur Cart Sidebar main item check karein.
4. "Leave Feedback" click karke rating aur comment submit karein.

Step 2 — Functional Checkout Demo

1. Cart sidebar se **"Proceed to Checkout"** click karein.
2. Shipping Address form, Delivery Speed (Standard/Express) select karein.
3. Promo Code main `FYP2026` enter karein (15% instant demo discount lag jayega).
4. **"Complete Order"** click karein → Order receipt page open hoga with unique Order Number and complete details.
5. Order directly MongoDB database main save ho jata hai.

Step 3 — Admin Intelligence Dashboard

1. Admin account se login karke `/admin` par jayen.
2. **Metrics Grid:** Total Customers, Total Orders, Demo Revenue, Total Feedback, Positive Sentiment % check karein.
3. **Gemini AI Cards:** "Generate Summary" aur "Actionable Recommendations" click karke live AI analysis show karein.
4. **Orders Database Table:** Show karein ke jo order customer ne place kiya tha, wo yahan real-time record ho chuka hai.

Step 4 — Customer 360 Journey (Deep Intelligence)

1. Orders Table ya Scatter Plot main kisi bhi customer par click karein.
2. Complete Customer 360 Timeline open hogi jisme customer ka pura lifecycle dikhayi dega:
PAGE VIEW → PRODUCT VIEW → ADD TO CART → ORDER PLACED (#ORD-XXXXX) .
3. Examiner ko show karein ke e-commerce aur analytics ka end-to-end loop 100% complete hai.